Learning a Hidden Partner

Bayesian vs. model-free accounts of trust updating in a repeated human-AI trust game

Project plan and model primer | Akshat Kakade | 5 Nov 2026 | Draft v0.1 | Context: CaCTüS 2027 application, Project 8 (computational RL models of human behaviour)

1. What I am trying to do

Fit and compare two simple computational models of how a person learns a **hidden partner's behaviour** over repeated rounds of a trust game, using a public human dataset. The deliverable is a short technical report and a small, tested code repository covering the whole pipeline: specify the models, simulate them, fit them to data, check that the fitting can be trusted (recovery), and compare them on held-out rounds.

Research question. Do participants' choices against a hidden AI partner look more like (A) model-free, reward-driven learning, or (B) Bayesian inference over the partner's behaviour?

Stated in advance. H1: the Bayesian model predicts held-out choices better than Q-learning, mostly in early rounds. H2: with about 20 rounds per task, parameters are only coarsely recoverable, and I will quantify how coarsely. Both models may fit about equally well (see Section 4.3); an indecisive result will be reported as indecisive.

2. Dataset and how it will be used

Source. Repeated trust-game experiments run at the Singapore University of Technology and Design (SUTD): Jayasekara, Prissé, Deng and Ho, *Exploring Trust in Artificial Intelligence (AI) Systems: Insights from a Repeated Trust Game* (SSRN 5229860; Journal of Behavioral and Experimental Economics, vol. 125, 2026). Per the abstract, participants played AI partners programmed with **social, competitive or exploratory** profiles and were given **no information** about the AI's objectives or incentives, so the partner's type is genuinely hidden from them. Data deposit: Mendeley Data 58mxtr44gm (the deposit I was pointed to; **not yet opened by me**). A sibling deposit from the same group, *Intrinsic Honesty and Trust in AI* (doi:10.17632/gjxztmgt37), describes three tasks: participant as trustor, participant as trustee, and changed endowments.

Day-1 go/no-go checklist (everything below is unconfirmed until checked)

Item to confirm	Why it matters
Rounds per task (reported as 20)	Sets how much data each person gives; limits what can be identified.
Participant role in each task (trustor or trustee)	Partner-learning applies cleanly when the participant is the trustor.
Per-round variables: amount sent, amount returned, endowments, multiplier	Defines the choice and the observation the models use.
Does the AI's move depend on the participant's choice?	Scripted vs. reactive partner changes what 'learning the partner' means.
Profile label per participant	Ground truth used only for analysis, never as a model input.
Risk / social-preference and AI-attitude questionnaires	Possible covariates; extension only.
Licence (reported CC BY 4.0) and citation requirement	Reuse and attribution.

How the data will be used

- **Unit of analysis:** one participant in one task, i.e. a sequence of (choice, outcome) pairs over rounds t = 1..T.
- **Models see only what the participant saw:** their own past choices and the amounts the partner returned. The AI profile label is used afterwards to ask whether fitted beliefs or learning rates differ by profile.
- **Choice coding:** discretise the amount sent into a few levels (e.g. none / low / mid / high) so the models stay tabular. (Assumption; to be adapted to the real choice format.)
- **Held-out split:** fit parameters on the first ~15 rounds of a task, then score the last ~5 with parameters frozen (beliefs and values still update through those rounds). Compare against a trivial baseline that always predicts the participant's most frequent choice.
- **Scope:** start with the task where the participant is the trustor; add others only if time allows. Data are public and anonymised; no attempt at re-identification; authors cited.

3. The task in model terms (stylised)

Each round the trustor holds an endowment E and sends an amount x. The amount is multiplied, and the partner returns an amount r. The trustor's payoff is E - x + r. The partner has a hidden tendency (how generously it returns), which the trustor can only infer from past returns. In POMDP language: the hidden state is the partner's behaviour, the observation is the returned amount, and the **belief** is the trustor's probability distribution over that

hidden behaviour. Exact multipliers and rules will be taken from the dataset documentation.

4. Model primer (intuition first, then the maths)

4.1 Softmax choice rule (shared by both models)

A model assigns each option a value V(a). Softmax turns values into choice probabilities:

$$P(a) = \exp(\beta \cdot V(a)) \,/\, \Sigma_b \exp(\beta \cdot V(b))$$

β is the *inverse temperature*: β → 0 means random choice, β → ∞ means always taking the highest-valued option. Intuition: people are mostly sensible but not deterministic, and a mistake becomes exponentially less likely the more it costs. Softmax also gives every choice a probability, which is what lets us compute a likelihood and fit the model to data.

4.2 Model 1: Q-learning (delta rule, model-free)

The agent keeps a value Q(a) for each option, meaning how well it has paid off so far. After choosing a and receiving reward r:

$$Q(a) \leftarrow Q(a) + \alpha \cdot (r - Q(a))$$

The bracket is the *prediction error* (reward minus expectation). α in [0, 1] is the *learning rate*: how far to move toward the latest outcome. The agent holds **no beliefs about the partner**; it only tracks which options paid off. Free parameters: α, β (and optionally the starting value Q_0).

4.3 Model 2: Bayesian partner inference

The agent keeps a **probability distribution over the partner's behaviour** and updates it with Bayes' rule. Let the partner's return fall into one of L levels (e.g. low / medium / high), with unknown probabilities $\theta = (\theta_1..\theta_L)$. Prior: Dirichlet($a_1..a_L$), where the a_j act as *pseudo-counts*. Because the Dirichlet is *conjugate* to the categorical distribution, updating is just counting: after n_j observations of level j,

$$\text{posterior} = \text{Dirichlet}(a + n), \quad P(\text{next return} = j) = (a_j + n_j) \,/\, \Sigma_k (a_k + n_k)$$

The agent then values each amount x by its expected utility, $EU(x) = \Sigma_j P(j) \cdot u(\text{payoff}(x, j))$, and chooses by softmax over EU. Free parameters: prior strength Σa_j (how much it trusts its prior vs. the data), prior mean (initially optimistic or pessimistic), risk curvature ρ in u, and β. The same utility transform u should be used in Model 1 so the comparison is about *learning*, not about taste for risk.

> **Key insight: the two models are closer than they look.** For one probability with prior pseudo-count total a_0 and n observations, the Bayesian posterior mean obeys
>
> $$m_n = m_{n-1} + (x_n - m_{n-1}) / (a_0 + n)$$
>
> which is exactly a delta rule whose learning rate is **$1/(a_0 + n)$**: large early, shrinking with experience. Q-learning uses a *constant* α. So the models mostly differ in how learning speed changes over rounds, and against a fixed scripted partner with ~20 rounds they may be hard to tell apart. That is why model recovery (Section 5) comes before any claim. Also note: Model 2 is *myopic*; it maximises this round's expected payoff and does not pay to gain information. Full Bayes-adaptive planning (a BAMDP agent that values information) is out of scope.

4.4 Fitting by maximum likelihood

The likelihood of a parameter set is the product, over rounds, of the probability the model gave to the choice the person actually made. We maximise its logarithm:

$$LL(\theta) = \Sigma_t \log P(a_t \mid \text{history}_t, \theta)$$

Intuition: how *unsurprised* is the model by what the person did? Models with more parameters always fit the training data better, so I compare them with penalised scores, AIC = 2k − 2·LL and BIC = k·ln(N) − 2·LL (k = number of parameters, N = number of choices), and, more importantly, by held-out log-likelihood.

4.5 Parameter recovery and model recovery

- **Parameter recovery:** simulate fake participants from known parameters, refit, and compare recovered with true values. If they do not match, parameter estimates from real people cannot be trusted.
- **Model recovery:** simulate from Model 1 and from Model 2, fit both models to every simulated dataset, and check that the better-fitting model is the true generator (a 2×2 confusion matrix). If the diagonal does not clearly dominate, the comparison is undecidable at this data size and I will say so.

4.6 Optional extension: shrinkage across people

With ~20 rounds each, individual estimates are noisy. Partial pooling pulls each person toward the group mean: $\theta_i = w \cdot \bar{x}_i + (1 - w) \cdot \mu$, with $w = \tau^2 / (\tau^2 + \sigma^2/n)$, where τ^2 is between-person spread and σ^2 is within-person noise. The noisier a person's data, the more the estimate leans on the group. Cut by default; included only if the core finishes early.

5. Evaluation plan

Check	Question it answers	Output
1. Parameter recovery	Can parameters be recovered at ~20 rounds?	True-vs-recovered scatter, correlations
2. Model recovery	Can the pipeline tell Q-learning from Bayesian inference?	2×2 confusion matrix
3. Model comparison (real data)	Which model fits participants better?	Per-person LL difference, AIC/BIC, count of people favouring each
4. Held-out prediction	Does a fitted model predict unseen rounds?	Held-out LL vs. most-frequent-choice baseline
5. Profile differences	Do fitted learning rates or priors differ by AI profile?	Parameters grouped by social / competitive / exploratory

6. Scope, fallback and schedule

In: two models, recovery checks, held-out prediction, short report, repo with unit tests (e.g. posterior counts, softmax sums to 1). **Out:** change-point models, IPOMDP / theory of mind, BAMDP planning, neural networks. **Fallback** if the dataset fails the Day-1 check: run the same pipeline on simulated agents playing the same stylised game, clearly labelled as simulation.

Date	Milestone
Nov 6	Data go/no-go checklist; descriptive statistics
Nov 7	Q-learning model and likelihood fitting
Nov 8	Bayesian partner-inference model
Nov 9	Model comparison on real data
Nov 10	Parameter and model recovery
Nov 11	Plots, short report, README
Nov 12	CV update and application statements
Nov 13	Submit (Nov 14-15 is buffer; endsems start Nov 16). Cut order if behind: held-out prediction first; recovery stays.

7. Limitations to state up front

- Partner policies are scripted, so 'learning the partner' is a simplified problem; results describe this dataset, not humans in general.
- About 20 rounds per task limits what any per-person fit can identify; the Bayesian model is myopic; choices also reflect fairness and risk preferences, not just beliefs.
- The dataset authors, per their abstract, focus on how partner personality shapes trust; this project asks a different question (learning dynamics) of the same data.
- Dataset details above are unconfirmed until the Day-1 check; this document will be updated with the real variable names and counts.

8. References

[1] Jayasekara, D., Prissé, B., Deng, R., Ho, J. Q. Exploring Trust in Artificial Intelligence (AI) Systems: Insights from a Repeated Trust Game. SSRN 5229860; J. Behav. Exp. Econ. 125 (2026).

[2] Jayasekara, D. et al. Intrinsic Honesty and Trust in AI (dataset). Mendeley Data, doi:10.17632/gjxztmgt37.

[3] Wilson, R. C., Collins, A. G. E. (2019). Ten simple rules for the computational modeling of behavioral data. eLife 8:e49547.

[4] Guez, A., Silver, D., Dayan, P. (2012). Efficient Bayes-Adaptive Reinforcement Learning using Sample-Based Search. NeurIPS.

[5] Shen, T., Dayan, P. Individual differences in tail risk sensitive exploration using Bayes-adaptive Markov decision processes. eLife (animal exploration data; BAMDP with a risk-sensitive objective).

[6] Hula, A., Montague, P. R., Dayan, P. Monte Carlo Planning method estimates planning horizons during interactive social exchange. arXiv:1502.03696.

[7] Alon, N., Schulz, L., Dayan, P., Barnby, J. M. (2023). Between Prudence and Paranoia: Theory of Mind Gone Right, and Wrong. ICML 2023 Workshop on Theory of Mind in Communicating Agents.